

MPhil Thesis: Literature Review

Mohammad-Shah Karim

May 2026

Word count: 1,099

Contents

1	Literature Review	2
1.1	Market Size and Pharmaceutical Innovation	2
1.2	Medicaid Expansion and Pharmaceutical Innovation	2
1.3	Medicaid Pricing and Manufacturer Incentives	3
1.4	Innovation Pipeline Timing	4

1 Literature Review

This thesis sits at the intersection of the literature on pharmaceutical innovation and how the institutional settings behind public reimbursement shape firm incentives. It also draws on work examining the timing of innovation measures across the drug-development pipeline

Before turning to these areas, it is beneficial to understand the current studies on the ACA Medicaid expansion. The reform yielded large gains in concentrated in states that chose to expand access to Medicaid, with the gaps in racial and ethnic coverage narrowing as a result (Courtemanche et al., 2020; Frean et al., 2017; Miller and Wherry, 2019). These states saw increased affordability indicating reductions in the cost barriers for accessing healthcare (Miller and Wherry, 2017), alongside declines in medical debt and bankruptcy from healthcare (Brevoort et al., 2017; Hu et al., 2018). These increases in access and financial welfare were accompanied with improvements in self-reported health and reductions in adult mortality (Miller et al., 2021; Miller and Wherry, 2017). However, evidence on labour market effects has been more limited, where no significant changes were recorded in employment outcomes or physician hours (Duggan et al., 2019; Dague et al., 2017; Neprash et al., 2021). This thesis studies a less explored dimension of the Medicaid expansion - its effects on pharmaceutical innovation.

1.1 Market Size and Pharmaceutical Innovation

1.2 Medicaid Expansion and Pharmaceutical Innovation

The most closely related study to this thesis is Garthwaite et al. (2021), which examines the impact of the ACA Medicaid expansion on clinical trial activity. With the use of disease condition Medicaid market shares as a proxy for Medicaid exposure, they find no evidence of a significant innovation response through to 2016. They propose that this null result may reflect the lower expected returns associated with Medicaid relative to reforms such as Medicare Part D, due to manufacturer revenue being constrained by the rebates and price controls embedded within the programme.

While this finding is an important starting point to address the question, their analysis leaves several gaps which this thesis addresses. First, the post-expansion window is constrained by both the innovation outcome studied and the length of the post-expansion window. Garthwaite et al. investigated clinical trial activity through to 2016, so their estimates only capture the first two years of the Medicaid demand shock. This is short relative to the development horizon for pharmaceutical drug development, which can span between five to twenty years (Brown et al., 2022). As a result, the absence of a response in clinical trials cannot distinguish between an innovation effect that is genuinely absent, or one that has affected early stage R&D decisions but has not reached the clinical trials stage.

Second, the absence of an aggregate clinical-trial response does not confirm whether the innovation effect differs across therapeutic classes. Sutton (1998) argues that pharmaceutical R&D is organised around narrow chemically related drug groups rather than broad therapeutic categories, and finds limited persistence in firms' major follow-on successes within these groups. This suggests that therapeutic-class-level aggregates may conceal meaningful heterogeneity in how firms adjust innovation activity. Instead, an insignifi-

cant aggregate relationship may be driven by heterogeneous responses across drug classes. Positive R&D responses may have occurred in classes where the market-size channel dominated, but the aggregate effect may have been offset by negative clinical-trial responses in areas where the expansion weakened innovation incentives.

Third, while Garthwaite et al. propose that the lower returns to the Medicaid market may explain the contrast with earlier market-size studies that find positive innovation responses, this explanation lies outside of their empirical design. Specifically, their analysis does not control for the procurement institutions that may affect manufacturer profits, including rebate pressures and formulary placement on Medicaid’s PDL.

This thesis addresses these gaps by extending the analysis across the innovation pipeline, across time, and across therapeutic markets. First, this thesis constructs novel panels linking firms to their drugs in Medicaid’s State Drug Utilisation Data, and then to their patenting and FDA approval activity. These innovation outcomes bracket the clinical-trial stage examined by Garthwaite et al., allowing earlier and later innovation responses to be considered separately. Second, the impact of the reform is investigated over a substantially longer post-expansion period, with outcome data extending to 2025, whereas Garthwaite et al. used data ending in 2016. This extension allows delayed innovation responses to be captured more clearly, which is particularly important given the long development horizon in pharmaceutical R&D. Third, class-level heterogeneous responses are examined to determine whether the expansion generated different innovation incentives across the therapeutic markets in which firms operated in, rather than restricting the analysis to an average therapeutic-area effect.

Garthwaite et al. cite the Medicaid procurement mechanism as a potential reason for the inconclusive results, thus this thesis addresses this paper directly in the empirical analysis with the use of firm-class exposure to Medicaid’s PDL. This is complemented by a counterfactual analysis of the clinical and pricing stages of preferred placement on the PDL, alongside a theoretical model in which market expansion interacts with rebates and placement on Medicaid’s formulary to shape innovation incentives.

1.3 Medicaid Pricing and Manufacturer Incentives

Medicaid prescriptions are subject to statutory rebates under the MDRP, and each state can negotiate supplemental rebates with manufacturers in exchange for preferred placement on the PDL. Duggan and Scott Morton (2006) show that these rules create incentives for manufacturers to raise prices for non-Medicaid consumers, since drug prices and reimbursement offered to Medicaid is linked to the prices charged in the private market. Consequently, the results highlight that manufacturers segment drug demand separately for the Medicaid and private markets, and the ACA may have intensified this distortion by raising the statutory rebate from 15.1% to 23.1% of AMP. Feng et al. (2023) find that the higher statutory rebates relaxed the constraints imposed by Medicaid’s most favoured customer clause allowing private customers to benefit with lower prices and lower drug spending by 2.5%. Together, these findings show that Medicaid’s rebate rules have spillover effects on pricing incentives outside of Medicaid, but the direction of the effect depends on the institutional channel through which the rebate structure operates.

Related structural work shows that pharmaceutical pricing rules can alter manufacturers’ pricing behaviour to offset reductions in per prescription revenues. Dubois et al. (2022)

show that firms adjust pricing and bargaining behaviour when price regulations in one market can impact profits elsewhere, indicating that the effect of pricing institutions on manufacturer returns depend on how firms respond strategically across markets.

However, Inderst and Wey (2007) show that larger buyers can raise incentives for incremental forms of product innovation, while weakening incentives for more substantial novel R&D. This distinction is crucial in the Medicaid setting, where the expansion increases market size, while also raising exposure to the formulary selection, which may weaken incentives for more inventive forms of innovation. Therefore, this is addressed empirically in the Medicaid setting by studying patent applications, which capture early stage R&D behaviour undertaken in anticipation of future returns. FDA approvals are used to assess whether any upstream innovation response is eventually reflected in regulatory outcomes.

1.4 Innovation Pipeline Timing

References

- Brevoort, K., Grodzicki, D., and Hackmann, M. B. (2017). Medicaid and financial health. Technical report, National Bureau of Economic Research.
- Brown, D. G., Wobst, H. J., Kapoor, A., Kenna, L. A., and Southall, N. (2022). Clinical development times for innovative drugs. *Nat Rev Drug Discov*, 21(11):793–794.
- Courtemanche, C., Marton, J., Ukert, B., Yelowitz, A., and Zapata, D. (2020). The impact of the affordable care act on health care access and self-assessed health in the trump era (2017-2018). *Health services research*, 55:841–850.
- Dague, L., DeLeire, T., and Leininger, L. (2017). The effect of public insurance coverage for childless adults on labor supply. *American Economic Journal: Economic Policy*, 9(2):124–154.
- Dubois, P., Gandhi, A., and Vasserman, S. (2022). Bargaining and international reference pricing in the pharmaceutical industry. Technical report, National Bureau of Economic Research.
- Duggan, M., Goda, G. S., and Jackson, E. (2019). The effects of the affordable care act on health insurance coverage and labor market outcomes. *National Tax Journal*, 72(2):261–322.
- Duggan, M. and Scott Morton, F. M. (2006). The distortionary effects of government procurement: evidence from medicaid prescription drug purchasing. *The Quarterly Journal of Economics*, 121(1):1–30.
- Feng, J., Hwang, T., and Maini, L. (2023). Profiting from most-favored-customer procurement rules: Evidence from medicaid. *American Economic Journal: Economic Policy*, 15(2):166–197.
- Frean, M., Gruber, J., and Sommers, B. D. (2017). Premium subsidies, the mandate, and medicaid expansion: Coverage effects of the affordable care act. *Journal of health economics*, 53:72–86.
- Garthwaite, C., Sachs, R., and Stern, A. D. (2021). Which markets (don’t) drive pharmaceutical innovation? evidence from u.s. medicaid expansions. Working Paper 28755, National Bureau of Economic Research.
- Hu, L., Kaestner, R., Mazumder, B., Miller, S., and Wong, A. (2018). The effect of the affordable care act medicaid expansions on financial wellbeing. *Journal of Public Economics*, 163:99–112.
- Inderst, R. and Wey, C. (2007). Buyer power and supplier incentives. *European Economic Review*, 51(3):647–667.
- Miller, S., Johnson, N., and Wherry, L. R. (2021). Medicaid and mortality: new evidence from linked survey and administrative data. *The Quarterly Journal of Economics*, 136(3):1783–1829.
- Miller, S. and Wherry, L. R. (2017). Health and access to care during the first 2 years of the aca medicaid expansions. *New England Journal of Medicine*, 376(10):947–956.

- Miller, S. and Wherry, L. R. (2019). Four years later: insurance coverage and access to care continue to diverge between aca medicaid expansion and non-expansion states. In *AEA Papers and Proceedings*, volume 109, pages 327–333. American Economic Association.
- Neprash, H. T., Zink, A., Sheridan, B., and Hempstead, K. (2021). The effect of medicaid expansion on medicaid participation, payer mix, and labor supply in primary care. *Journal of Health Economics*, 80:102541.
- Sutton, J. (1998). *Technology and market structure : theory and history*. The MIT Press, Cambridge, Mass, 1st ed. edition.